Design of a Communication System for an Early Warning System at a Nickel Mining Site: A LoRaWAN-Based Approach with Fault-Tolerant and Adaptive Transmission Strategies

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***Abstract*—This study is conducted as part of a computer network course at the Department of Computer Science and Electronics, Universitas Gadjah Mada, and is intended primarily as a learning-based design and recommendation rather than a fully validated implementation. The work is based on existing technical research and focuses on the design of a reliable and fault-tolerant communication architecture for a landslide Early Warning System (EWS) in a mining environment. The paper analyses key aspects of the communication design, including link budget estimation, spreading factor trade-offs, routing and recovery mechanisms, data buffering, and delay-tolerant operation.**

Although the system has not yet been deployed or experimentally validated, this work provides a technically grounded reference and practical recommendation for designing reliable communication systems in remote environments.

***Index Terms*—LoRaWAN, Early Warning System, Landslide, Nickel Mining, Mesh Topology, Fault Tolerance, IoT, MQTT, Synchronous Communication, Asynchronous Communication, GSM Fallback.**

I. Introduction

Landslides represent one of the most significant hazards at open-pit nickel mining sites. Heavy rainfall, altered slope geometry from excavation, and unstable soil structures can trigger rapid mass movement with little warning, putting personnel and equipment at risk. This work originates from direct field observation at an active nickel mining site where landslide events have occurred unexpectedly and, in some cases, resulted in casualties.

Most published research on landslide Early Warning Systems (EWS) focuses on the *sensor layer*: which tilt meters, piezometers, or rainfall gauges to use, and what threshold values to set [4], [5]. This paper addresses the *communication layer* instead, because reliable data delivery from field sensors to operators is equally critical yet receives far less attention in the literature, especially for remote industrial environments.

The communication system must satisfy three core requirements: (1) reliable operation across several kilometres of rugged, metal-dense terrain; (2) delivery of critical alerts within a defined time budget even when primary internet is unavailable; and (3) deployment and maintenance by a small on-site team.

II. Communication Topology and Protocol Selection

A. Why Mesh Topology?

Three standard topologies were evaluated: star, bus, and mesh. Table I summarises their trade-offs for a mining-site EWS. Mesh is selected because each node can relay packets for its neighbours, so the network survives even when individual nodes are buried or destroyed by a landslide event.

TABLE I: Network Topology Comparison for Mining-Site EWS

Criterion	Star	Bus	Mesh
Node failure tolerance	Low	Low	**High**
Link failure tolerance	Low	Very Low	**High**
Coverage flexibility	Moderate	Low	**High**
Configuration effort	Simple	Simple	Moderate
Energy per node	Low	Low	Moderate
Suitable for EWS	Partial	No	**Yes**

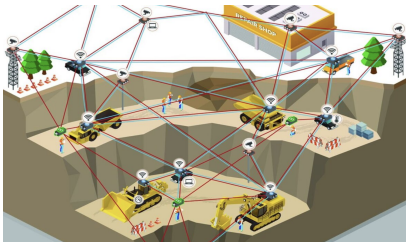

Fig. 1: Mesh Topology

B. LoRaWAN Protocol Selection

LoRa (Long Range) modulation is used for the wireless links between sensor nodes and gateways. It operates in the sub-GHz ISM band, for Indonesia, the 920–923 MHz frequency range (subject to KOMINFO regulation). LoRaWAN Class A is selected for sensor nodes, giving good power efficiency: a node wakes briefly to transmit, then goes back to sleep.

The link budget tells us whether a radio signal is strong enough to be received reliably at a given distance. The key formula is:

$$P_{rx} = P_{tx} + G_{tx} + G_{rx} - L_{fs} \tag{1}$$

The received signal power (P_{rx}) represents the strength of the signal at the gateway after it has traveled through the wireless channel. It is determined by the transmitted power (P_{tx}), the gains of both the transmitting and receiving antennas (G_{tx} and G_{rx}), and the signal attenuation over distance, known as free-space path loss (L_{fs}). In general, the signal starts with a certain power at the transmitter, is slightly amplified by the antennas, and then gradually weakens as it propagates through space. The amount of signal loss increases with both distance and frequency. This means that longer communication distances and higher operating frequencies result in greater attenuation.

LoRa communication also depends on a parameter called the Spreading Factor (SF), which determines the trade-off between communication range and data rate. A lower SF enables faster data transmission but limits the communication range, while a higher SF allows signals to travel longer distances at the cost of reduced data rate. Protocol comparison across candidate technologies is provided in Table II. LoRaWAN is the clear choice for multi-kilometre outdoor sensor networks.

TABLE II: Communication Protocol Comparison for Mining-Site EWS

Parameter	**LoRa**	**Zigbee**	**Wi-Fi**	**RS-485**
Range (km)	2–10	0.1–0.3	0.05–0.15	1.2 (wired)
Data rate (kbps)	0.3–50	250	54,000+	10,000
Power TX (mW)	25–100	60–100	300–600	N/A
Signal penetration	Excellent	Moderate	Poor	N/A
Deployment type	Wireless	Wireless	Wireless	Wired
ISM band (MHz)	868/915	2400	2400/5000	N/A
Suitable for EWS	**Yes**	Partial	No	Limited

III. Fault Tolerance and Redundancy

A. How the Mesh Handles Node Failures

When a sensor node loses its primary communication link, for example due to being buried by a landslide, the network must automatically establish an alternative route to maintain connectivity. In this system, the AODV (Ad-hoc On-demand Distance Vector) routing protocol is used to handle such situations in a dynamic manner.

In practice, when a link failure occurs, the affected node initiates a route discovery process by broadcasting a Route Request (RREQ) message to its neighbouring nodes. These neighbouring nodes then forward the request through the network until a valid path to the gateway is found. Once a reachable node or the gateway itself responds with a Route Reply (RREP), a new communication path is established and data transmission can resume.

B. Gateway Redundancy and Packet Protection

Multiple RAK7249 gateways are deployed at different positions across the site with at least 15% overlap in their coverage zones, so every sensor node is within range of at least two gateways simultaneously. If one gateway fails, the other automatically takes over without any manual intervention.

At the packet level, LoRa uses Forward Error Correction (FEC): extra bits are added to every packet so that some bit errors during transmission can be automatically corrected without retransmission. A 16-bit Cyclic Redundancy Check (CRC) is also appended to every packet to detect any remaining errors. The principle is:

$$\text{CRC} = \text{remainder}\left(\frac{\text{Message}}{\text{Generator}}\right) \tag{2}$$

The sender computes a short number from the data and attaches it. The receiver recomputes the same number and compares, if they don't match, the packet was corrupted and is discarded. FEC goes further by actually fixing certain errors before they're even checked.

IV. Data Transmission Strategy

A. Dual-Mode: Periodic and Event-Triggered

The system uses two transmission modes running simultaneously, as shown in Fig. 2. **Periodic** packets go out every 10–15 minutes for baseline monitoring; **alert** packets are sent immediately when a threshold is crossed.

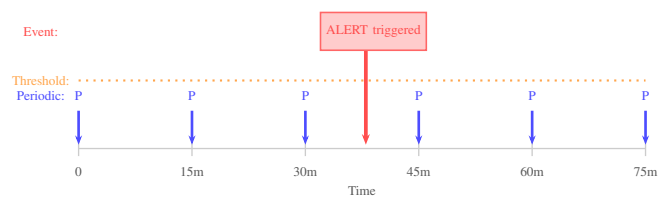

Fig. 2: Dual-mode transmission timeline. Blue arrows show periodic packets every 15 minutes; the red arrow shows an immediate alert packet triggered when sensor data crosses the threshold at ≈38 minutes.

The packet structure is designed to support both periodic monitoring and alert-based transmissions with a compact and efficient format. Each packet includes essential fields such as Node ID, timestamp, soil tilt, vibration, moisture level, battery status, and system flags, which are present in both periodic and alert messages. For alert transmissions, additional fields are included, namely the alert type and diagnostic metadata, to provide more detailed information about abnormal conditions. As a result, periodic packets have a total size of approximately 50 bytes, while alert packets are slightly larger at around 70 bytes due to the extra diagnostic content.

TABLE III: Geotechnical Alert Thresholds (Site-Specific Calibration Required)

Parameter	**Unit**	**Warning**	**Critical**
Soil tilt angle	degrees	> 3	> 5
Vibration amplitude	g	> 0.05	> 0.1
Moisture increase rate	%/10 min	> 5	> 10
Cumulative rainfall	mm/hr	> 25	> 50

V. Hardware Architecture

TABLE IV: Hardware and Power Summary

Component	Key Specifications
Sensor Node	SX1276 LoRa (-148 dBm), STM32L4 (80 MHz), LiPo + solar, $< 10\,\mu$A sleep, months of operation
Gateway	RAK7249 (8-channel), 4G LTE, IP67, GPS, 16 GB SSD, 5 dBi antenna
Solar Support	20 W panel $\approx$ 80–100 Wh/day, sufficient for continuous operation

The selected hardware components are chosen to balance performance, reliability, and energy efficiency for long-term field deployment. The SX1276 LoRa transceiver is used due to its high sensitivity, enabling long-range communication even in challenging terrain. The STM32L4 microcontroller is selected for its low power consumption makes sensor nodes to operate for months using a combination of battery and solar energy. On the gateway side, the RAK7249 provides robust connectivity with 4G LTE support, local data storage, and weather-resistant design, suitable for outdoor environments. The use of solar power for sustainable operation and reducing the need for frequent maintenance in remote areas.

VI. Design Adaptation for Unstable Internet Connectivity

A. Data Buffering and Delay-Tolerant Operation

The system implements delay-tolerant operation by storing data locally during connectivity outages. The gateway uses a 16 GB SSD, where critical alerts are in a protected partition, while routine data is managed using a circular buffer. At the node level, available flash memory allows storage of approximately 1800 packets, equivalent to about 19 days of data at 15-minute intervals. When the connection is restored, the gateway checks link quality before uploading buffered data using sequence numbers to maintain correct order, with retries managed through exponential backoff to prevent network congestion.

B. Adaptive Transmission Under Variable Bandwidth

The gateway monitors backhaul quality continuously and adjusts its behaviour automatically, as shown in Table V.

TABLE V: Adaptive Transmission Strategy Based on Backhaul Quality

Bandwidth	Behaviour	Compression
>100 kbps	Full upload (routine + alert), 15 min	None
50–100 kbps	Routine compressed, 15 min interval	$\approx$40% (zlib)
10–50 kbps	Routine throttled to 1 hr, alerts full	$\approx$40% (zlib)
<10 kbps	Alerts only (uncompressed), routine paused	None (alerts)
No connection	Fallback: GSM/SMS or 433 MHz radio	Not applicable

Alert packets are *never* compressed to avoid any added latency in the time-critical delivery path.

C. Priority Levels for Data Transmission

Three priority levels apply consistently from node-to-gateway and gateway-to-server, illustrated in Fig. **??**.

TABLE VI: Three-Tier Data Priority Scheme

Priority	Level	Description
P1	High	Critical geotechnical alerts; MQTT QoS 2, up to 5 retries (10 s interval), allocated up to 90% bandwidth
P2	Medium	Node health data (battery, RSSI); transmitted every 30 minutes using QoS 1
P3	Low	Routine sensor data every 15 minutes; QoS 0, paused when P1 is active or bandwidth <10 kbps

D. Fallback Alert Delivery Under Internet Failure

When an alert is triggered by a sensor, the system attempts to deliver it through a prioritized sequence of communication channels to reliability. The process begins by checking internet availability, if available, the alert is sent using MQTT QoS 2 with up to five retries at 10-second intervals. If the internet is unavailable, the system falls back to GSM by sending an SMS message via a SIM800L module, with up to ten retries at one minute intervals. If GSM is also unavailable, the system attempts to transmit the alert using a 433 MHz radio link to a nearby base station. If all communication methods fail, a local siren and LED beacon are activated after a 15-minute delay to provide a physical warning on-site.

The SMS alert message follows a compact 160-character format, for example: `[ALERT] Zone:A Tilt:6.2deg Vibr:0.12g 09:45 UTC+7`.

VII. Synchronous and Asynchronous Communication Modes

A. Key Difference

Synchronous: The sender waits for an explicit acknowledgment before proceeding. If no reply comes in time, it retries. **Asynchronous:** The sender transmits and immediately moves on without waiting; delivery is verified later.

Fig. 3 illustrates this conceptual difference.

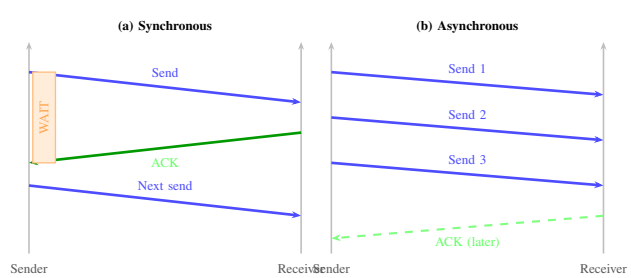

Fig. 3: Conceptual comparison of (a) synchronous and (b) asynchronous communication. In synchronous mode the sender blocks (orange) until acknowledged; in asynchronous mode multiple messages are sent without waiting, with a deferred acknowledgment.

B. Application to Both Scenarios

In a stable network (Scenario 1), synchronous communication using MQTT QoS 2 provides the highest delivery reliability through acknowledgment and retries, but with higher

bandwidth and power consumption. In contrast, asynchronous communication uses QoS 0 or QoS 1 to transmit data without waiting for acknowledgments for faster processing and higher throughput, supported by local buffering. In unstable network conditions (Scenario 2), synchronous communication remains reliable but requires longer timeouts and may fall back to SMS or radio links when internet connectivity is unavailable. Meanwhile, asynchronous communication stores data locally and transmits it when connectivity is restored, while still allowing immediate delivery of critical alerts through the fastest available channel.

Overall, asynchronous mode is more efficient for routine data, while synchronous mode is best reserved for critical alerts. Therefore, a hybrid approach is recommended, combining asynchronous transmission for regular monitoring data with synchronous, acknowledged delivery for high-priority alerts.

VIII. Detailed Hardware, Software, and Configuration Specification

A. Hardware Configuration Summary

The system uses a consistent hardware platform across both scenarios, with additional components introduced in Scenario 2 to improve reliability under unstable network conditions. All sensor nodes are equipped with the SX1276 LoRa transceiver and STM32L4 microcontroller, powered by a LiPo battery with solar charging for long-term operation. The gateway is based on the RAK7249, which provides multi-channel LoRaWAN support, GPS, and local data storage. In Scenario 1, communication relies on a stable LTE or fiber backhaul. In Scenario 2, additional fallback mechanisms are included, such as GSM and 433 MHz radio, to maintain connectivity during outages.

- **Core hardware (both scenarios):** SX1276 LoRa, STM32L4 MCU, LiPo + solar, RAK7249 gateway, 16 GB storage
- **Scenario 1:** Dedicated LTE/fiber backhaul only
- **Scenario 2:** Adds GSM (SIM800L) and 433 MHz radio as fallback
- **Antenna:** 5 dBi LoRa + dipole (radio fallback)
- **Power:** 12 V battery with 20 W solar panel

B. System Performance Summary

Fig. 4 shows the modelled alert delivery latency vs. backhaul packet loss for all four operating configurations.

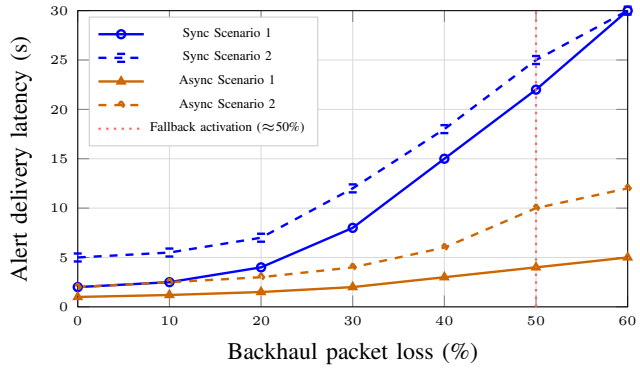

Fig. 4: Estimated alert delivery time under different levels of network packet loss. Asynchronous communication (orange) keeps the delay low even when the network quality decreases. In contrast, synchronous communication (blue) becomes slower as packet loss increases, but provides more reliable delivery. In Scenario 2, backup communication methods (GSM or radio) are activated when packet loss reaches around 50%.

IX. Conclusion

This paper presents a LoRaWAN-based communication system design for a landslide early warning system (EWS) in a mining environment, with a focus on communication reliability. The use of a mesh topology and LoRaWAN enables long-range, low-power communication suitable for distributed sensor networks. Link budget analysis shows a strong margin at 5 km, indicating that the system can operate reliably even in challenging conditions.

The proposed design combines periodic monitoring with event-based alerts, allowing both continuous observation and fast response when needed. Two deployment scenarios are considered, along with fallback mechanisms such as GSM/SMS, radio communication, and local alarms to handle network failures.

Overall, the results suggest that asynchronous communication is more efficient for routine data, while synchronous communication is more suitable for critical alerts. Although this work has not yet been experimentally validated, it provides a practical reference for designing communication systems in remote and high-risk environments.

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